

OptiCrop – Smart Agricultural Crop Recommendation System Using Machine Learning

Project Title

OptiCrop – Smart Agricultural Crop Recommendation System Using Machine Learning

Abstract

OptiCrop is a Machine Learning-based Smart Agricultural Crop Recommendation System developed to assist farmers and agricultural professionals in selecting the most suitable crop based on soil nutrients and climatic conditions. Agriculture plays a vital role in the economy, and choosing the right crop is essential for improving productivity, reducing financial risks, and promoting sustainable farming practices. Traditional crop selection methods mainly depend on farmers' experience and local knowledge, which may not always produce the best results under changing environmental conditions.

The OptiCrop system uses Machine Learning techniques to analyze important agricultural parameters such as Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, Soil pH, and Rainfall. These parameters are processed using a trained Logistic Regression model to predict the most suitable crop for cultivation.

The trained Logistic Regression model achieved an overall prediction accuracy of **98.30%**, demonstrating the effectiveness of Machine Learning in solving realworld agricultural problems. By combining data analysis, predictive modeling, and web development, OptiCrop provides a practical solution that supports farmers in making informed crop selection decisions.

Problem Statement

Agriculture is one of the most important sectors for food production and economic development. Selecting the right crop is a challenging task because it depends on several factors such as soil nutrients, temperature, humidity, rainfall, and soil PH. Traditional crop selection methods often rely on experience and may not always produce the best results, leading to low productivity and financial losses.

The objective of this project is to develop a Machine Learning-based web application that recommends the most suitable crop based on soil and environmental conditions. The system uses a Logistic Regression model to analyze input parameters including Nitrogen (N), Phosphorous (P), Potassium (K), Temperature, Humidity, pH, and Rainfall. It provides quick and accurate crop recommendations through a simple Flask-based web interface, helping farmers make better agricultural decisions and improve crop productivity.

Features

- Predicts the most suitable crop using Machine Learning.
- Uses the Logistic Regression algorithm for accurate crop recommendations.
- Interactive Flask web application with a simple and user-friendly interface.
- Accepts seven important Agricultural Parameters as input.
 - 1 .Nitrogen (N)
 - 2 .Phosphorous (P)
 - 3 .Potassium (K)
 - 4 .Temperature
 - 5 .Humidity
 - 6 . Soil pH
 - 7 Rainfall
- ☐ Displays instant crop prediction results.
- ☐ Responsive web interface designed using HTML and CSS.
- ☐ Easy deployment using the Render cloud platform.
- ☐ Well-organized GitHub repository with complete project documentation.

Achieved an overall prediction accuracy of **98.30%**.

Tech Stack

Programming Language

- Python 3.11

Machine Learning Libraries

- Pandas
- NumPy
- Scikit-learn
- Pickle

Web Framework

- Flask

Frontend Technologies

- HTML5
- CSS3 Development

Tools

- Visual Studio Code
- Git
- GitHub

Deployment Platform

- Render
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Installation Steps

Step 1

Clone the repository:

```
git clone https://github.com/sk34567/OptiCrop-AI-ML-Project.git
```

[Step](#)

2

Navigate to the project directory:

```
cd OptiCrop-AI-ML-Project
```

[Step](#)

3

Install all required dependencies:

```
pip install -r requirements.txt
```

[Step](#)

4

Run the Flask application:

```
python app.py
```

Step 5

Open your browser and visit:

<http://127.0.0.1:5000> or access the

deployed application:

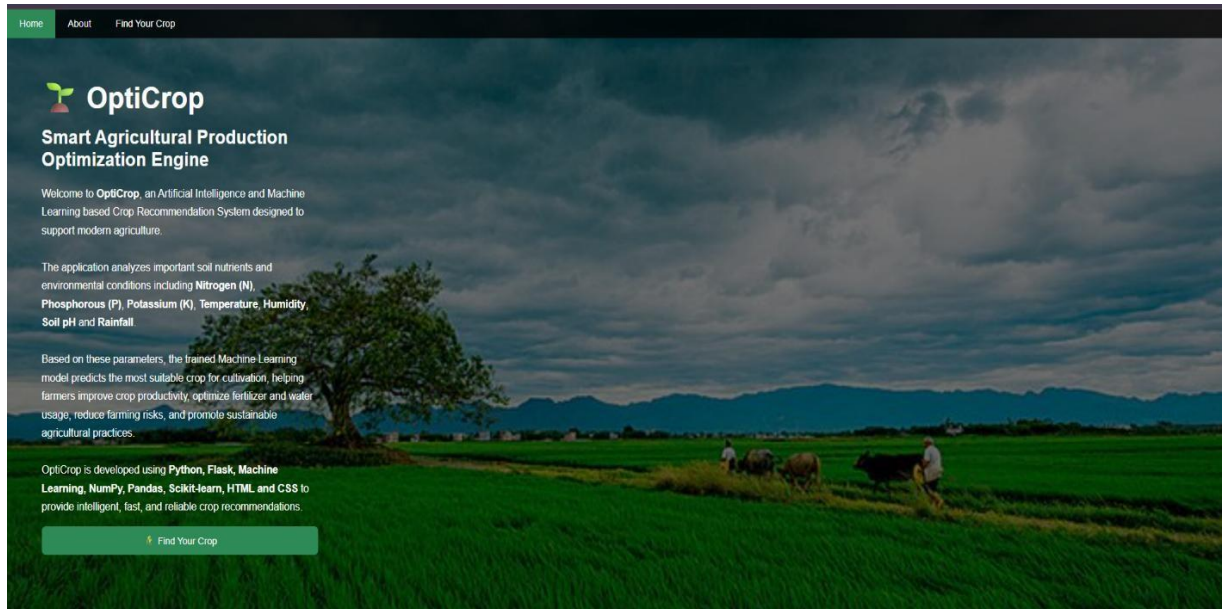
<https://opticrop-ai-ml-project.onrender.com>

Usage Instructions

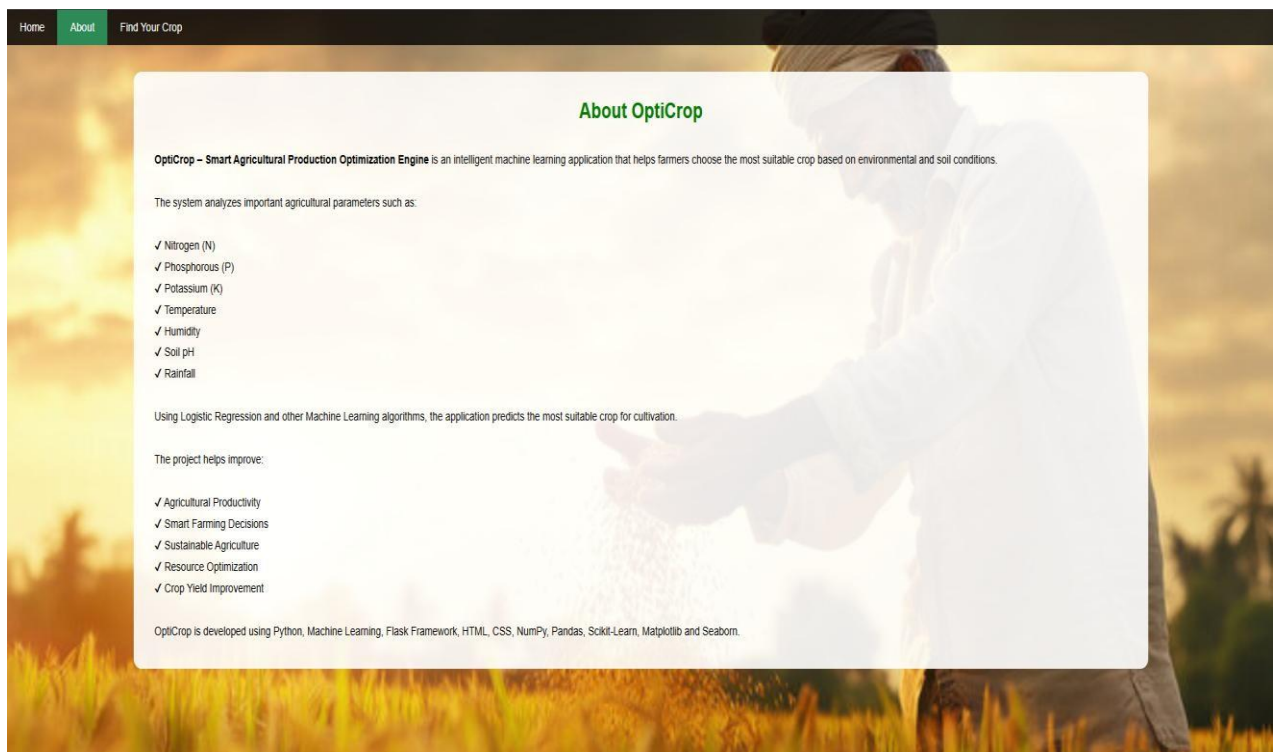
1. Open the **OptiCrop** web application.
 2. Enter the following input values:
 - ☐ Nitrogen (N)
 - ☐ Phosphorous (P)
 - ☐ Potassium (K)
 - ☐ Temperature
 - ☐ Humidity
 - ☐ Soil pH
 - ☐ Rainfall
 3. Click the **Predict** button.
 4. The Machine Learning model processes the input values
 5. The application displays the **recommended crop** based on the given soil and climatic conditions.
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Screenshots

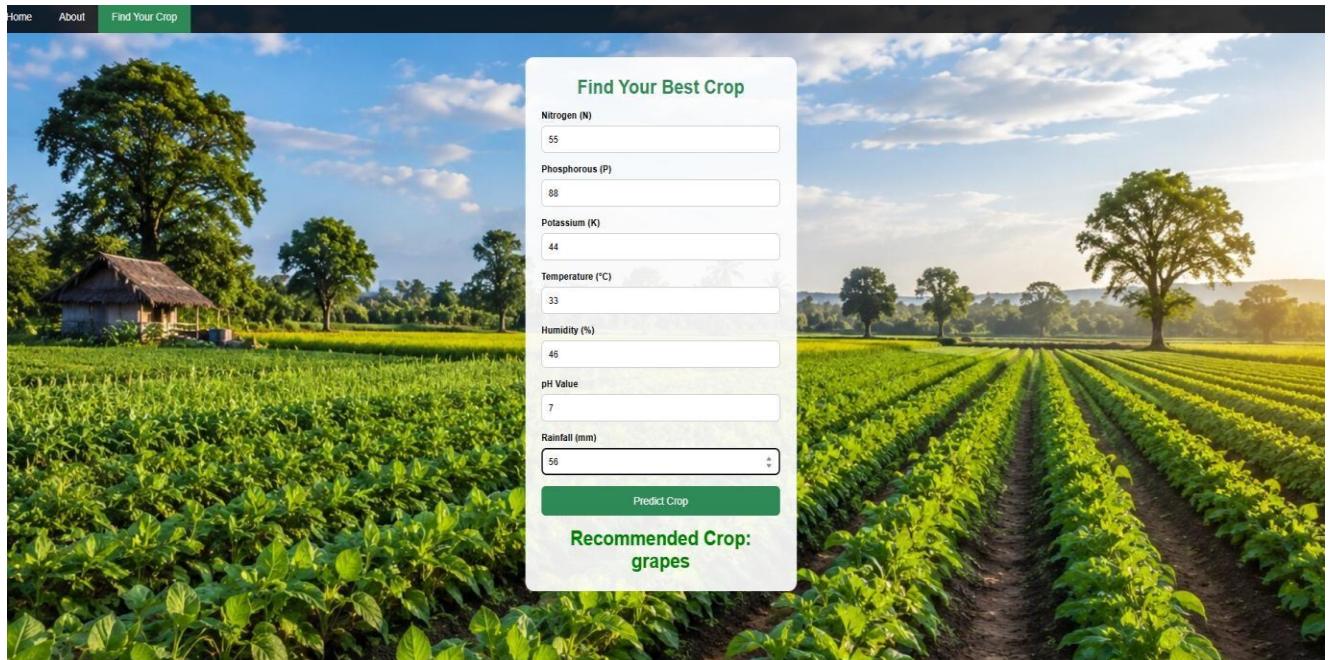
1. Home Page



2. About Page



3.Find Your Crop Page



The screenshot shows a web application interface for finding the best crop. The background is a lush green vineyard with rows of grapevines under a blue sky with scattered clouds. A small white building is visible on the left. Overlaid on the center is a white card titled "Find Your Best Crop". The card contains several input fields for agricultural data: Nitrogen (N) with value 55, Phosphorous (P) with value 88, Potassium (K) with value 44, Temperature (°C) with value 33, Humidity (%) with value 45, pH Value with value 7, and Rainfall (mm) with value 56. Below these fields is a green button labeled "Predict Crop". At the bottom of the card, it displays "Recommended Crop: grapes" in green text. A navigation bar at the top of the page has three links: "Home", "About", and "Find Your Crop", with "Find Your Crop" being the active link.

Parameter	Value
Nitrogen (N)	55
Phosphorous (P)	88
Potassium (K)	44
Temperature (°C)	33
Humidity (%)	45
pH Value	7
Rainfall (mm)	56

Predict Crop

Recommended Crop: grapes

Future Scope

Although the current application provides accurate crop recommendations, several enhancements can be implemented in future versions:

- ☐ Recommend suitable fertilizers based on soil conditions
- ☐ Provide soil health analysis and improvement suggestions.
- ☐ Support multiple regional and international languages.
- ☐ Develop a mobile application for Android and iOS users.
- ☐ Store user history and crop recommendations in a cloud database.
- ☐ Compare the performance of different Machine Learning algorithms.
- ☐ Integrate satellite and IoT sensor data for precision agriculture.
- Deploy the application using scalable cloud services for better performance.

Team Member Details

S.No	Name	Role
1	Aki Harsha Vardhini	Machine Learning Developer & Flask Application Developer (Team Lead)
2	Yamparala Sravani	Machine Learning Developer & Flask Application Developer (Team Member)
3	Venkata Pavana Lakshmi Santa Javvaji	Machine Learning Developer & Flask Application Developer (Team Member)
4	Shaik Yasin	Machine Learning Developer & Flask Application Developer (Team Member)
5	Kosuru Lokesh	Machine Learning Developer & Flask Application Developer (Team Member)